

Lorenzo Poggioni

PhD Candidate in Applied Mathematics & Scientific Computing
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■ Research Interests

Numerical Analysis, High-Performance Computing (HPC), Computational Fluid Dynamics (CFD), High-order numerical methods, Nonlinear Wave Propagation.

■ Publications & Preprints

- 2025** **L. Poggioni**, D. Clamond & Y. D'Angelo & S. Abide, “A novel infinite-order method for nonlinear advection models: The zigzag schemes.” *hal-05075534* [math.NA]. (Under Review).
- In Prep.** **L. Poggioni**, “A robust infinite-order method for compressible Navier–Stokes equations.”
- In Prep.** **L. Poggioni**, “Construction and implementation of optimised dealiasing kernels for pseudo-spectral methods.”

■ Education

- 2023 –** **PhD Candidate**, Numerical Modeling and Fluid Dynamics group, J.A. Dieudonné Lab, Nice, France.
Thesis: High-order numerical methods for PDEs. Supervised by Didier Clamond.
- 2022–2023** **Master 2, “Mathématiques Pures et Appliquées” (Honors)**, Université Côte d’Azur.
Specialization in Theoretical and Applied Mathematics.
- 2020–2022** **Master 1 & last year of Bachelor**, Université Côte d’Azur, France.
Mathematics curriculum, specialised in scientific computing.
- 2018–2020** **Classes préparatoires aux Grandes Écoles (CPGE)**, Valbonne, France.
Intensive program in Mathematics, Physics and Computer Science (MPSI/MP).
- 2015–2018** **Baccalauréat scientifique**, Lycée Guillaume Apollinaire, France.
Programming specialty (ISN).

■ Research Experience

- Nov 2025** **Research Guest**, Technological University Dublin, Ireland. (Host: Rossen Ivanov).
Mathematical modeling of internal water waves in layered media.
- 2023** **Research Internship**, Lagrange Lab, Observatory Côte d’Azur, France.
Developed novel spectral methods for nonlinear conservation laws.
- 2022** **Research internship**, J.A. Dieudonné Lab, Nice, France.
Studied numerical methods for the propagation of unstable flame fronts.

■ Technical Skills

- Programming** Fortran, Matlab, Python (NumPy, SciPy), LaTeX.
- HPC** Parallel computing with MPI and CUDA.
- Languages** French (Native), English (Fluent), Japanese (JLPT N4).

■ Academic Awards

Dec 2025 **Best Poster Award**, French Group of Rheology (GFR).
Awarded at the Fluids and Complexity conference.

■ Teaching Experience

2023–2025 **Teaching Assistant**, Université Côte d’Azur.
192 hours of Bachelor-level instruction: Linear Algebra, Probability, Statistics.

2023–2024 **Colleur (Oral Examiner)**, Université Côte d’Azur.
Oral examinations in Linear Algebra II for Bachelor students.

■ Talks & Conferences

June 2026 **GdR Mécanique des Fluides Géophysiques et Astrophysiques & Navier–Stokes 2.00**, Toulouse, France.
Talk about a new high-order finite difference based methods with improved stability and robustness for computational fluid mechanics.

Dec 2025 **Fluids and Complexity**, Nice, France.
Poster showcasing results about an improved spectral method for nonlinear advection models.

Nov 2025 **Technological University Dublin**, Dublin, Ireland.
Talk about a new hybrid finite difference scheme featuring improved stability constraints for solving advection-dominated problems.

Nov 2025 **Laboratory’s Ph.D. Seminar**, Nice, France.
Talk about the construction of new finite difference schemes with arbitrary stencils.

June 2025 **Mat du Labo**, Nice, France.
Poster showcasing Ph.D. results.

May 2025 **Applied Analysis Nice–Toulon–Marseille day**, Porquerolles, France.
Talk about an improved finite difference scheme for linear advection problems.

Nov 2024 **Laboratory’s Ph.D. Seminar**, Nice, France.
Talk about advanced tools for numerical integration.

May 2024 **Analysis workshop of the laboratory**, Nice, France.
Talk about an introduction to variational principles in physics.

March 2024 **Math PhD Students Colloquium**, Les Cévennes, France.
Talk about an introduction to numerical integration and adaptive time stepping.

June 2023 **Ulyseus Spring School in PDEs**, Sevilla, Spain.